

Mengran Li

PhD Candidate in Statistics

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Summary

I develop causal inference methodology designed for rare and extreme events. My work contributes to causal discovery, estimation, and attribution in settings involving extreme events and extremal dependence. By synthesizing causal frameworks with extreme value theory, I aim to enable reliable inference in high-impact extreme regimes with a focus on climate risk assessment.

Education

University of Glasgow

Doctor of Philosophy in Statistics

Glasgow, UK

2022 - 2026

University of Glasgow

Master of Science in Statistics with distinction

Glasgow, UK

2021 - 2022

Southwestern University of Finance and Economics

Bachelor of Science in Statistics

Bachelor of Economics in Finance (Double Degree)

Chengdu, China

2015 - 2019

Publications

* denotes corresponding author

1. **Li, M.*** & Castro-Camilo, D. (2026) Causal Discovery in Multivariate Extremes via Tail Asymmetry. *arXiv:2604.21620*. Under review at *JASA (Theory & Methods)*.
2. **Li, M.*** & Castro-Camilo, D. (2026). Tail-Calibrated Estimation of Extreme Quantile Treatment Effects. *arXiv:2603.23309*. Under review at *JRSS-B*.
3. **Li, M.*** & Castro-Camilo, D. (2025). On the importance of tail assumptions in climate extreme event attribution. *arXiv:2507.14019*. Revision invited at *Climatic Change*.
4. **Li, M.***, Cuba, D., Hu, C., & Castro-Camilo, D. (2024). A wee exploration of techniques for risk assessments of extreme events. *Extremes*. doi:10.1007/s10687-024-00500-5
5. **Li, M.*** & Castro-Camilo, D. (2025). eFCM: An R package for the Exponential Factor Copula Model. CRAN R package. cran.r-project.org/package=eFCM

Research Experience

Causal Structure Learning for Multivariate Extremes

Developed structure learning methods for high-dimensional multivariate extremes and tail dependence, enabling recovery of directional causal structure encoded by directed acyclic graphs in extreme regimes.

Glasgow, UK

2025 - 2026

Causal Inference for Extreme Quantile Treatment Effects

Developed a causal inference framework based on inverse estimating equations for estimating treatment effects in the far tails of outcome distributions, with formal identification and asymptotic guarantees.

Glasgow, UK

2023 - 2025

Tail Dependence and Causal Attribution of Climate Extremes

Integrated multivariate extreme value modeling into a counterfactual causal inference framework to assess the role of tail dependence in extreme event attribution, showing that dependence assumptions critically shape causal conclusions.

Glasgow, UK

2022 - 2025

Modeling Extreme Quantiles and Multivariate Extremes

Co-developed a modeling strategy for extreme quantiles and multivariate extremes, integrating extreme value and dependence modeling techniques. Results published in *Extremes*.

Glasgow, UK

2023 - 2024

Predictive Modeling for Extreme Event Attribution

Motivated by the EVA Data Challenge 2025, developed ConvLSTM-based predictive models for attribution and risk estimation of extreme events, capturing spatio-temporal dependence and tail behaviour; now an independent methodological study with manuscript in preparation.

Glasgow, UK

2025 - 2026

Teaching Experience

University of Glasgow

Tutor and Demonstrator

Glasgow, UK

2022 – 2026

Statistics & Modeling Bayesian Statistics, Time Series, Flexible Regression, Biostatistics, Advanced Predictive Models

Statistical Computing Introduction to R & Python, Data Analysis

Responsibilities Led tutorials and lab sessions for undergraduate and MSc students; provided technical support for programming assignments; assessed coursework and provided constructive feedback.

Conference Presentations

1. **First workshop of the Glasgow-Edinburgh Extremes Network**, 11 December 2025, University of Glasgow, Glasgow (UK). *Talk presentation.*
2. **ISI World Statistics Congress 2025**, 5–9 October 2025, International Statistical Institute, The Hague (Netherlands). *Talk presentation.*
3. **RSS 2025 International Conference**, 1–4 September 2025, Royal Statistical Society, Edinburgh (UK). *Poster presentation.*
4. **The 6th International Conference on Advances in Extreme Value Analysis and Application to Natural Hazards (EVAN 2024)**, 16–19 July 2024, Istituto Veneto di Scienze, Venice (Italy). *Talk presentation.*
5. **STOR-i Extremes Workshop**, 20–22 September 2023, Lancaster University, Lancaster (UK). *Talk presentation.*
6. **Research Students’ Conference in Probability and Statistics**, 11–14 September 2023, University of Sheffield, Sheffield (UK). *Talk presentation; Travel grant recipient.*

Awards and Recognitions

- **CDSAI Training and Conference Fund (Competitive Award).** University of Glasgow, 2026.
- **Google Cloud Research Grant.** December 2025.
- **3rd Place, IASC Data Analysis Competition, Travel grant awarded.** ISI World Statistics Congress, The Hague (Netherlands), July 2025.
- **Fully funded PhD scholarship (CSC).** 2022–2026.
- **Academic Scholarships.** Southwestern University of Finance and Economics. 2015–2019.

Academic Service

Journal Reviewing

2025 - Present

Journal reviewer: *Data-Centric Engineering* (Cambridge University Press);
Co-reviewer: *Annals of Applied Statistics* (with supervisor).

Seminar & Reading Group Organisation

2025 - Present

Co-organized the monthly **Glasgow–Edinburgh Extremes Network (GLE²N)** seminar series, coordinating invited speakers and supporting seminar delivery for sessions hosted at the University of Glasgow.

Skills

Programming & Computing	R (advanced), Python, C++, Linux, Git, HPC, Cloud Computing (Google Cloud)
Scientific Software & Reproducibility	R package development (CRAN), R Markdown, LaTeX
Web & Documentation	HTML/CSS, Hugo